

Stat 122 (Mathematical Statistics II)

Problem Set No. 7

INSTRUCTIONS: Answer the following as indicated. Show detailed solutions.

1. Suppose that the random variable Y has a gamma distribution with parameters $\alpha = 2$ and an unknown β .
 - a. Show that $Q = 2Y/\beta$ is a pivotal quantity. What is the distribution of Q ?
 - b. Use Q to construct a 90% confidence interval for β .
2. A random sample of 130 healthy human body temperatures yielded a mean 98.25 degrees and standard deviation 0.73 degrees.
 - a. Construct and interpret a 99% confidence interval for the average body temperature of healthy people.
 - b. Does the confidence interval obtained in part (a) contain the value 98.6 degrees, the accepted average temperature cited by physicians and others? What conclusions can you draw?
3. For a comparison of the rates of defectives produced by two assembly lines, independent random samples of 100 items were selected from each line. Line A yielded 18 defectives in the sample, and line B yielded 12 defectives.
 - a. Find a 98% confidence interval for the true difference in proportions of defectives for the two lines.
 - b. Is there evidence here to suggest that one line produces a higher proportion of defectives than the other?
4. Organic chemists often purify organic compounds by a method known as fractional crystallization. An experimenter wanted to prepare and purify 4.85 g of aniline. Ten 4.85-gram specimens of aniline were prepared and purified to produce acetanilide. The following dry yields were obtained: 3.85, 3.88, 3.90, 3.62, 3.72, 3.80, 3.85, 3.36, 4.01, 3.82. Construct and interpret a 95% confidence interval for the mean number of grams of acetanilide that can be recovered from 4.85 grams of aniline.

5. The ages of a random sample of five university professors are 39, 54, 61, 72, and 59. Using this information, find a 99% confidence interval for the population standard deviation of the ages of all professors at the university, assuming that the ages of university professors are normally distributed.
6. A state wildlife service wants to estimate the mean number of days that each licensed hunter actually hunts during a given season, with a bound on the error of estimation equal to 2 hunting days. If data collected in earlier surveys have shown σ to be approximately equal to 10, how many hunters must be included in the survey?